

15. A set of garden lights is powered by an array of solar cells.

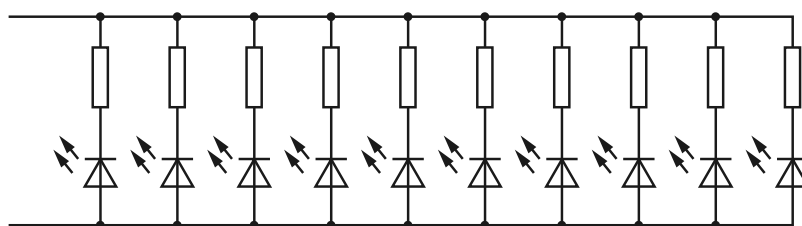


The set of garden lights consists of 10 LEDs connected in parallel.

Each LED is connected in series with a resistor.

The combined resistance of each LED and resistor is $220\ \Omega$.

Part of the circuit is shown.



- (a) Calculate the total resistance of this part of the circuit.

3

Space for working and answer



* X 8 5 7 7 6 0 1 4 6 *

15. (continued)

- (b) A solar cell is a p-n junction.

When photons of light are incident on the p-n junction, a potential difference is produced.

- (i) State the name of this effect.

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- (ii) Using band theory, explain how a potential difference is produced when photons of light are incident on the p-n junction.

3

[Turn over



* X 8 5 7 7 6 0 1 4 7 *